Solutions: Algebra and Analysis, IMC 2001–2006

March 2026

Algebra

Solution 1. (IMC 2003, Day 2, Problem 1)
Since $AB + A + B = 0$, we can write

$$AB + A + B + I = I \implies (A + I)(B + I) = I.$$

So $A + I$ and $B + I$ are inverses of each other. Then

$$(B + I)(A + I) = I$$

as well. Expanding:

$$BA + B + A + I = I \implies BA + B + A = 0 = AB + A + B,$$

hence $AB = BA$. □

Solution 2. (IMC 2003, Day 1, Problem 3)
The minimal polynomial of A divides $3x^3 - x^2 - x - 1$. Checking $x = 1$: $3 - 1 - 1 - 1 = 0$, so

$$3x^3 - x^2 - x - 1 = (x - 1)(3x^2 + 2x + 1).$$

The discriminant of $3x^2 + 2x + 1$ is $4 - 12 = -8 < 0$, so its roots are $\frac{-1\pm i\sqrt{2}}{3}$, each with modulus $\sqrt{\frac{1+2}{9}} = \frac{1}{\sqrt{3}} < 1$.

Since the polynomial $3x^3 - x^2 - x - 1$ has three distinct roots, the minimal polynomial has distinct roots, so A is diagonalizable: $A = C^{-1}DC$, where D is diagonal with entries from $\{1, \frac{-1+i\sqrt{2}}{3}, \frac{-1-i\sqrt{2}}{3}\}$.

Since the two non-real eigenvalues have modulus $\frac{1}{\sqrt{3}} < 1$, we have $D^k \to M$ where M is diagonal with 1's on positions corresponding to eigenvalue 1 and 0's elsewhere. Then $M^2 = M$ and

$$\lim_{k\to\infty} A^k = C^{-1}MC,$$

which is idempotent: $(C^{-1}MC)^2 = C^{-1}M^2C = C^{-1}MC$. □

Solution 3. (IMC 2001, Day 1, Problem 5)

We prove by induction on n. For $n = 2$, let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $b \neq 0$, conjugating by $\begin{pmatrix} 1 & 0 \\ a/b & 1 \end{pmatrix}$ gives a matrix with 0 in the $(1, 1)$ position. Similarly for $c \neq 0$. If $b = c = 0$ and $a \neq d$, conjugating by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ gives $\begin{pmatrix} a & d-a \\ 0 & d \end{pmatrix}$, reducing to the case $b \neq 0$.

For the inductive step $n > 2$: write $A = \begin{pmatrix} A' & * \\ * & \beta \end{pmatrix}_n$. If $A' \neq \lambda I$, apply induction to get $P^{-1}A'P = \begin{pmatrix} 0 & * \\ * & \alpha \end{pmatrix}_{n-1}$. Conjugating A by $\begin{pmatrix} P^{-1} & 0 \\ 0 & 1 \end{pmatrix}$, the diagonal becomes $(0, 0, \ldots, 0, \alpha, \beta)$. View the lower-right 2×2 block and apply the $n = 2$ idea to reduce to at most one non-zero diagonal entry. (Full details require careful case analysis when $A' = \lambda I$, which can be handled by permuting rows/columns to find a submatrix $\neq \lambda I$.) □

Solution 4. (IMC 2002, Day 2, Problem 5)
$\Leftarrow$: If $A = S\overline{S}^{-1}$, then $A\overline{A} = S\overline{S}^{-1} \cdot \overline{S}S^{-1} = I_n$. ✓
$\Rightarrow$: Choose $S = wA + \overline{w}I_n$ for a complex number w. Then

$$A\overline{S} = A(\overline{w}\,\overline{A} + wI) = \overline{w}A\overline{A} + wA = \overline{w}I + wA = S.$$

So $A = S\overline{S}^{-1}$ whenever S is invertible. Now S is singular iff $-\overline{w}/w$ is an eigenvalue of A. Since A has finitely many eigenvalues and $\overline{w}/w$ can be any complex number on the unit circle, we can choose w so that S is invertible. □

Solution 5. (IMC 2002, Day 2, Problem 1)

Add the second row to the first, add the third to the second, $\dots$, add the n-th row to the $(n-1)$-th. The first $n-1$ rows become $(1,1,0,\dots,0)$, $(0,1,1,0,\dots,0)$, etc. Now subtract the first column from the second, the resulting second from the third, and so on. After these operations:

$$\det(A) = \begin{vmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & n+1 \end{vmatrix} = n+1.$$

□

Solution 6. (IMC 2005, Day 1, Problem 1)
Since $a_{ij} = i + j$, the matrix A can be written as

$$A = \mathbf{v}\mathbf{e}^T + \mathbf{e}\mathbf{v}^T,$$

where $\mathbf{v} = (1, 2, \dots, n)^T$ and $\mathbf{e} = (1, 1, \dots, 1)^T$. So A is a sum of two rank-1 matrices, hence $\operatorname{rank}(A) \le 2$.

For $n = 1$, the rank is 1 (the matrix is (2)).

For $n \ge 2$, the top-left 2×2 minor is $\begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = 8 - 9 = -1 \ne 0$, so $\operatorname{rank}(A) \ge 2$.

Therefore $\operatorname{rank}(A) = 2$ for $n \ge 2$, and $\operatorname{rank}(A) = 1$ for $n = 1$. □

Solution 7. (IMC 2005, Day 2, Problem 3)
Answer: $\dfrac{n(n-1)}{2}$.

Upper bound: If A is a nonzero symmetric matrix, then $\operatorname{trace}(A^2) = \operatorname{trace}(A^T A) = \sum a_{ij}^2 > 0$, so V contains no nonzero symmetric matrix. The space of symmetric matrices has dimension $\frac{n(n+1)}{2}$, so $V \cap S = \{0\}$ and

$$\dim V \le n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

Attainability: The space of strictly upper triangular matrices has dimension $\frac{n(n-1)}{2}$. For any two strictly upper triangular matrices X, Y, the product XY is also strictly upper triangular, so all diagonal entries of XY are zero, giving $\operatorname{trace}(XY) = 0$. □

Solution 8. (IMC 2006, Day 1, Problem 3)
By induction, it suffices to prove the case $k = 2$: if $\det A = bc$, find integer matrices B, C with $\det B = b$, $\det C = c$, $A = BC$.

WLOG A is upper triangular (multiply by unimodular matrices from left/right). For upper triangular A, define B_{nn} to be the gcd of A_{nn} and b, set $C_{nn} = A_{nn}/B_{nn}$. Then construct the remaining entries of B and C inductively from row n upwards, solving the linear equations $A_{i,n} = B_{i,j}C_{j,n} + \cdots + B_{i,n}C_{n,n}$. The coprimality conditions ensure integer solutions exist at each step. □

Solution 9. (IMC 2006, Day 2, Problem 6)
If some $A_j = \lambda I$, then $A_j = B_j$ (from the product condition), and the result is trivial.

Otherwise, some A_j, say A_3, has two distinct real eigenvalues λ, μ. We may assume $A_3 = B_3 = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$ by conjugating. Let $A_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B_2 = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$.

From $A_1A_2A_3 = I = B_1B_2B_3$: $\operatorname{trace}(A_2A_3) = \operatorname{trace}(A_1^{-1}) = \operatorname{trace}(B_1^{-1}) = \operatorname{trace}(B_2B_3)$, giving $a\lambda + d\mu = a'\lambda + d'\mu$. Similarly using other trace identities, we get $a = a'$, $d = d'$, and $bc = b'c'$.

If $b = 0$ or $c = 0$, then $(1,0)^T$ or $(0,1)^T$ would be a common eigenvector of all A_i, contradiction. So $b, c \ne 0$. Take $S = \begin{pmatrix} c' & 0 \\ 0 & c \end{pmatrix}$ (up to a scalar). Then S commutes with $A_3 = B_3$ and conjugates A_2 to B_2, so it conjugates $A_1 = (A_2A_3)^{-1}$ to $B_1 = (B_2B_3)^{-1}$. □

Solution 10. (IMC 2004, Day 2, Problem 1)

Let $A = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$ and $B = (B_1 \; B_2)$ where A_1, A_2, B_1, B_2 are 2×2 matrices. Then

$$AB = \begin{pmatrix} A_1B_1 & A_1B_2 \\ A_2B_1 & A_2B_2 \end{pmatrix} = \begin{pmatrix} I_2 & -I_2 \\ -I_2 & I_2 \end{pmatrix}.$$

So $A_1B_1 = I_2$, $A_1B_2 = -I_2$, $A_2B_1 = -I_2$, $A_2B_2 = I_2$.

From $A_1B_1 = I_2$: $B_1 = A_1^{-1}$. From $A_1B_2 = -I_2$: $B_2 = -A_1^{-1}$. Then $A_2 = -B_1^{-1} = -A_1$.

Therefore:

$$BA = B_1A_1 + B_2A_2 = A_1^{-1}A_1 + (-A_1^{-1})(-A_1) = I_2 + I_2 = \boxed{2I_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}}.$$

$\square$

Analysis

Solution 11. (IMC 2001, Day 1, Problem 3)

Set $t = e^{-h}$ where $h \to 0^+$. Then $1 - t \sim h$ and

$$(1-t)\sum_{n=1}^{\infty} \frac{t^n}{1+t^n} = \frac{1-t}{-\ln t} \cdot (-\ln t) \sum_{n=1}^{\infty} \frac{t^n}{1+t^n}.$$

Since $\frac{1-t}{-\ln t} \to 1$ as $t \to 1$, it suffices to compute

$$\lim_{h \to 0+} h \sum_{n=1}^{\infty} \frac{1}{1+e^{nh}}.$$

This is a Riemann sum for the integral

$$\int_0^{\infty} \frac{dx}{1+e^x} = \left[-\ln(1+e^{-x})\right]_0^{\infty} = 0 - (-\ln 2) = \ln 2.$$

□

Solution 12. (IMC 2001, Day 2, Problem 5)

Suppose such f exists. If $f(f(x)) \le 0$ for all x, then f is decreasing (take $y \le 0$: $f(x+y) \ge f(x) + yf(f(x)) \ge f(x)$). Since $f(0) > 0 \ge f(f(x))$, we get $f(x) > 0$ for all x, contradiction.

So there exists z with $f(f(z)) > 0$. The inequality $f(z+x) \ge f(z) + xf(f(z))$ shows $\lim_{x\to\infty} f(x) = +\infty$ and hence $\lim_{x\to\infty} f(f(x)) = +\infty$.

There exist $x, y > 0$ with $f(x) \ge 0$, $f(f(x)) > 1$, $y \ge \frac{x+1}{f(f(x))-1}$, so $f(x+y+1) \ge 0$. Then

$$\begin{aligned} f(f(x+y)) &\ge f(x+y+1) + (f(x+y) - (x+y+1))f(f(x+y+1)) \\ &\ge f(x) + yf(f(x)) + f(f(x+y)) > f(f(x+y)), \end{aligned}$$

a contradiction. □

Solution 13. (IMC 2002, Day 1, Problem 2)

No. Suppose such f exists. Since $f'(x) = f(f(x)) > 0$, f is strictly monotone increasing. By monotonicity, $f(f(x)) > f(0)$ for all x, so $f'(x) > f(0)$.

Then for all $x < 0$:

$$f(x) = f(0) + \int_0^x f'(t)\,dt < f(0) + x \cdot f(0) = (1+x)f(0).$$

For $x \le -1$, this gives $f(x) \le 0$, contradicting $f(x) > 0$. □

Solution 14. (IMC 2003, Day 1, Problem 5)

Answer: $\lim_{n\to\infty} f_n(x) = g(0)$ for every $x \in (0, 1]$.

For non-decreasing g: by induction, each f_n is non-decreasing, and $g(x) = f_0(x) \ge f_1(x) \ge \cdots \ge g(0)$. The limit $h(x) = \lim f_n(x)$ exists and $h \ge g(0)$.

If $h(1) > g(0)$, choose δ with $h(1) > g(\delta)$. Then $f_{n+1}(1) \le \delta g(\delta) + (1-\delta)f_n(1)$, so $f_n(1) - f_{n+1}(1) \ge \delta(f_n(1) - g(\delta)) \ge \delta(h(1) - g(\delta)) > 0$, and $f_n(1) \to -\infty$, contradiction. So $h(1) = g(0)$, hence $h \equiv g(0)$.

For general continuous g: let $M(x) = \sup_{[0,x]} g$ (non-decreasing) and $m(x) = \inf_{[0,x]} g$ (non-increasing). Define M_n, m_n analogously. By induction $m_n(x) \le f_n(x) \le M_n(x)$. By the monotone case, both $M_n(x) \to M(0) = g(0)$ and $m_n(x) \to m(0) = g(0)$. By squeezing, $f_n(x) \to g(0)$. □

Solution 15. (IMC 2003, Day 2, Problem 2)

For $x \in (0, \frac{\pi}{2})$ and $t \in [x, 2x]$, we use $\frac{\sin t}{t}$ is decreasing on $(0, \pi)$ and $\lim_{t\to 0} \frac{\sin t}{t} = 1$. Then

$$\left(\frac{\sin 2x}{2x}\right)^m \int_x^{2x} \frac{t^m}{t^n}\,dt < \int_x^{2x} \frac{\sin^m t}{t^n}\,dt < \int_x^{2x} \frac{t^m}{t^n}\,dt,$$

and

$$\int_x^{2x} t^{m-n}\,dt = x^{m-n+1} \int_1^2 u^{m-n}\,du.$$

The factor $\left(\frac{\sin 2x}{2x}\right)^m \to 1$. So the limit depends on x^{m-n+1}:

$$\lim_{x\to 0+} \int_x^{2x} \frac{\sin^m t}{t^n}\, dt = \begin{cases} 0, & \text{if } m \geq n \text{ (i.e. } m-n+1>0), \\ \ln 2, & \text{if } n-m=1, \\ +\infty, & \text{if } n-m>1. \end{cases}$$

□

Solution 16. (IMC 2004, Day 1, Problem 3)

a) The set $Y = \{(y_1, \ldots, y_n) : 0 \leq y_i \leq 1, \sum y_i = 1\} \subset \mathbb{R}^n$ is connected. The function $f(y) = \arcsin y_1 + \cdots + \arcsin y_n$ is continuous, so $S_n = f(Y)$ is a connected subset of $\mathbb{R}$, hence an interval.

b) Since $\sin x$ is concave on $[0, \frac{\pi}{2}]$, by Jensen's inequality:

$$\frac{1}{n} \sin \frac{x_1 + \cdots + x_n}{n} \geq \frac{\sin x_1 + \cdots + \sin x_n}{n} \cdot \frac{1}{1} \implies \frac{2}{\pi}(x_1 + \cdots + x_n) \leq 1.$$

Actually more precisely: $\frac{2}{\pi} \sum x_k \leq \sum \sin x_k = 1$ gives $\sum x_k \leq \frac{\pi}{2}$.

The minimum is achieved when all $x_i = \arcsin \frac{1}{n}$, giving $\sum x_k = n \arcsin \frac{1}{n}$.

So $S_n = \left[n \arcsin \frac{1}{n}, \frac{\pi}{2}\right]$ and

$$l_n = \frac{\pi}{2} - n \arcsin \frac{1}{n} \to \frac{\pi}{2} - \lim_{n\to\infty} \frac{\arcsin(1/n)}{1/n} = \frac{\pi}{2} - 1.$$

□

Solution 17. (IMC 2004, Day 2, Problem 5)

Solution 1. Use $x^{-1} - 1 \geq |\ln x|$ for $x \in (0, 1]$ (since both sides equal 0 at $x = 1$ and $(x^{-1} - 1)' = -x^{-2} \leq -x^{-1} = |\ln x|'$ on $(0, 1)$). Therefore

$$\int_0^1 \int_0^1 \frac{dx\, dy}{x^{-1} + |\ln y| - 1} \leq \int_0^1 \int_0^1 \frac{dx\, dy}{|\ln x| + |\ln y|}.$$

Substituting $y = u/x$:

$$\int_0^1 \int_0^1 \frac{dx\, dy}{|\ln x| + |\ln y|} = \int_0^1 \frac{1}{|\ln u|} \cdot |\ln u|\, du = 1.$$

(Detailed computation: substitute $s = x^{-1} - 1$, $u = s - \ln y$ and use Fubini.) □

Solution 2. Substituting $s = x^{-1} - 1$ and $u = -\ln y$:

$$\int_0^1 \int_0^1 \frac{dx\, dy}{x^{-1} + |\ln y| - 1} = \int_0^\infty \int_s^\infty \frac{e^{s-u}}{(s+1)^2 \cdot u}\, du\, ds = \int_0^\infty \left(\int_0^u \frac{e^s}{(s+1)^2}\, ds \right) \frac{e^{-u}}{u}\, du.$$

By convexity of $\frac{e^s}{(s+1)^2}$:

$$\int_0^u \frac{e^s}{(s+1)^2}\, ds \leq \frac{u}{2} \left(\frac{e^u}{(u+1)^2} + 1 \right),$$

giving

$$\leq \frac{1}{2} \left(\int_0^\infty \frac{du}{(u+1)^2} + \int_0^\infty e^{-u}\, du \right) = \frac{1}{2}(1+1) = 1.$$

□

Solution 18. (IMC 2005, Day 1, Problem 3)

Let $M = \max_{0 \leq x \leq 1} |f'(x)|$. Since $-M \leq f'(x) \leq M$, integrating:

$$-M \int_0^x f(t)\, dt \leq \frac{1}{2} f^2(x) - \frac{1}{2} f^2(0) \leq M \int_0^x f(t)\, dt.$$

Multiplying by $f(x) \geq 0$ and integrating over $[0, 1]$:

$$-Mf(x) \int_0^x f(t)\, dt \leq \frac{1}{2} f^3(x) - \frac{1}{2} f^2(0) f(x) \leq Mf(x) \int_0^x f(t)\, dt.$$

Integrating the rightmost inequality over $[0,1]$:

$$\int_0^1 f^3 - f^2(0)\int_0^1 f \le 2M\int_0^1 f(x)\int_0^x f(t)\,dt\,dx = M\left(\int_0^1 f\right)^2.$$

Similarly for the left inequality, giving the desired bound. $\square$

Solution 19. (IMC 2005, Day 2, Problem 4)
Define

$$g(x) = -\frac{f(-1)}{2}x^2(x-1) - f(0)(x^2-1) + \frac{f(1)}{2}x^2(x+1) - f'(0)x(x-1)(x+1).$$

One can verify that $g(\pm 1) = f(\pm 1)$, $g(0) = f(0)$, and $g'(0) = f'(0)$.

Apply Rolle's theorem to $h(x) = f(x) - g(x)$: since $h(-1) = h(0) = h(1) = 0$, there exist $\eta \in (-1,0)$, $\vartheta \in (0,1)$ with $h'(\eta) = h'(\vartheta) = 0$. Also $h'(0) = 0$, so h' has three zeros. By Rolle's theorem applied twice more, h'' has two zeros, then h''' has a zero $\xi \in (-1,1)$.

Computing:

$$g'''(x) = -3f(-1) + 3f(1) - 6f'(0) = 3(f(1) - f(-1)) - 6f'(0).$$

So $0 = h'''(\xi) = f'''(\xi) - g'''(\xi)$, giving

$$f'''(\xi) = 3(f(1) - f(-1)) - 6f'(0), \quad \frac{f'''(\xi)}{6} = \frac{f(1) - f(-1)}{2} - f'(0).$$

$\square$

Solution 20. (IMC 2006, Day 2, Problem 2)
The functions $f(x) = x + c$ and $f(x) = -x + c$ clearly satisfy the condition. We prove these are the only ones.

Since $f([a,b])$ is a closed interval of length $b-a$, we have $|f(x) - f(y)| \le |x-y|$ for all x, y (the image can't exceed the length). So f is continuous.

Given $x < y$, let $a, b \in [x,y]$ achieve the max and min of f on $[x,y]$. Then $f(\max) - f(\min) = y - x$, but also $|a-b| \le y - x$, so

$$y - x = f(a) - f(b) \le |a-b| \le y - x.$$

Hence $\{a,b\} = \{x,y\}$ and f is monotone. If f is increasing: $f(x) - f(y) = x - y$ implies $f(x) - x = f(y) - y$, so $f(x) = x + c$. Similarly, if decreasing, $f(x) = -x + c$. $\square$